

Curriculum Vitae

Changyo Han

Ph.D. in Information Science and Technology

HCI Researcher

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🎓 <https://scholar.google.com/citations?user=jvNYnSIAAAAJ>

Employment

- 2024 – ···· 📌 **Project Lecturer**, The University of Tokyo
- 2021 – 2024 📌 **Assistant Professor**, The University of Tokyo
- 2020 – 2021 📌 **JSPS Postdoctoral Research Fellow**, The University of Tokyo
- 2019 – 2020 📌 **JSPS Research Fellowship (DC2)**, The University of Tokyo
- 2013 – 2016 📌 **Researcher**, Electronics and Telecommunications Research Institute (ETRI)

Education

- 2017 – 2020 📌 **Ph.D. in Information Science and Technology, The University of Tokyo, Japan**
Thesis title: *Force Markers: Embossed Fiducials for Recognizing Physical Objects on Pressure-Sensitive Touch Surfaces.*
- 2011 – 2013 📌 **M.E., The University of Tokyo, Japan.**
- 2007 – 2011 📌 **B.E., Tokyo Institute of Technology, Japan**

Grants

- 2021–2025 📌 **JSPS Grant-in-Aid for Early-Career Scientists**
- 2020–2023 📌 **JST ACT-X Frontier of mathematics and information science**
- 2019–2021 📌 **JSPS Research Fellowship for Young Scientists (DC2, PD)**

Honors and Awards

Fellowships

- 2019–2021  **JSPS Research Fellowship for Young Scientists (DC2, PD)**
- 2017–2020  **Graduate Program for Social ICT Global Creative Leaders**, The University of Tokyo.
- 2012–2013  **Seiho scholarship**, Seiho scholarship foundation
- 2006–2011  **Japan-Korea Joint Program for Science and Engineering**,
Full government scholarship both from Korea and Japan.

Awards

- 2024  **Interactive Presentation Award**, IEICE HCG Symposium 2024.
-  **Best Poster Award**, IMPS 2024.
-  **h.a.n.d. Corporation Award**, EC 2024.
-  **Excellent Teaser Video Award**, EC 2024.
-  **Interactive Presentation Award (audience vote)**, IPSJ Interaction 2024.
-  **Interactive Presentation Award (audience vote)**, IPSJ Interaction 2024.
- 2023  **Best Poster Award**, IMPS 2023.
-  **Interactive Presentation Award (audience vote)**, IPSJ Interaction 2023.
- 2022  **FIT Encouragement Award**, FIT 2022.
- 2021  **MVE Award**, IEICE MVE 2021.
-  **CHI'21 People's Choice Best Demo Honorable Mention Award** (2nd place), CHI 2021.
- 2020  **Honorable Mention Award** (top 5%), CHI 2020.
-  **Interactive Presentation Award (PC recommendation)**, IPSJ Interaction 2020.
-  **Interactive Presentation Award (audience vote)**, IPSJ Interaction 2020.
-  **Best Paper Award** (top 1 paper), TEI 2020.

Teaching Experience

Lecture

- 2021  UTokyo iii-Exhibition, **The University of Tokyo**.
<https://www.iiiexhibition.com/>

Teaching Assistant

- 2011  Student experiment on optical communication, **The University of Tokyo**.

Teaching Experience (continued)

- Lectures on electromagnetism II, **The University of Tokyo**.

Mentoring

- 2025
 - **Yuchi Yahagi, Yasunori Akashi**, PhD thesis.
 - **Yutaro Konishi, Jingfan Xi**, Master thesis.
 - **Kota Murakami, Misaki Kyono, Mizuki Ogawa**, Bachelor thesis.
- 2024
 - **Taichi Morimura, Yasunori Akashi**, Master thesis.
 - **Ryoma Kanai, Keita Inoue**, Bachelor thesis.
- 2023
 - **Yuki Kakui, Yosuke Nakagawa**, Master thesis.
 - **Yasunori Akashi**, Bachelor thesis.
- 2022
 - **Yuchi Yahagi**, Master thesis.
 - **Minami Aramaki, Kota Araki**, Bachelor thesis.
- 2021
 - **Ikuo Kamei**, Master thesis.
 - **Yuki Kakui, Ryotaro Tanaka**, Bachelor thesis.
- 2020
 - **Ryo Takahashi**, Master thesis.
 - **Yuchi Yahagi**, Bachelor thesis.
- 2018
 - **Saho Yamaguchi**, Bachelor thesis.

Academic Service

Organizer

- Secretary Member
 - Virtual Reality Society in Japan Art+Entertainment Research Committee
- Conferences
 - VRST '22 Local Arrangements/Online Chair, TEI '23 Associate Chair

Reviewer

- CHI '21–'25, VRST '21, UIST '20, TEI '20–'24, DIS '24, SCF '23, SIGGRAPH Asia '24, MobileHCI '21, Audio Mostly '20
- Journals
 - Trans. HI Society, IEEE/Optica JOCN, IEEE/Optica Optics Express, Trans. VRSJ

Exhibitions, Workshops

- 2023 –  **AIR-range Plus** (UTokyo iii-Exhibition 2023)
<https://iii-exhibition2023-main.vercel.app/>
- 2019 –  **Musical instrument interface design with tangible markers and pressure-sensitive sensor array** (GCL/GDWS Workshop C)
<https://gcl-gdws.org/workshop-c/2019/musical-instrument-interface-design-with-tangible-markers-and-pressure-sensitive-sensor-array/>
- 2018 –  **Inventory management system using BumpMarker** (CEATEC Japan 2018 JAPERABooth), ,
<https://www.nedo.go.jp/content/100925847.pdf>

Miscellaneous Experience

Invited Talks

- 2024-05-29  **Design and Fabrication of Adaptive Physical Interfaces for Users** Lecture for Department of Engineering and Design, Faculty of Engineering and Design, Hosei University – Introduction to System Design
- 2023-11-25  **For Those Hosting a Workshop for the First Time in Their Life** The Virtual Reality Society of Japan, Art + Entertainment Research Committee – A+E Meetup-3
<https://sigae.vrsj.org/?p=502>
- 2021-06-01  **The Evolution of Tangible User Interfaces** The University of Tokyo – Introduction to Human-Computer Interaction
- 2020-04-28  **PneuModule: Using Inflatable Pin Arrays for Reconfigurable Physical Controls on Pressure-Sensitive Touch Surfaces** CHI 2020 Japan Chapter local meeting
<https://sigchi.jp/event/chi2020-japan-chapter/>

Panel Talks

- 2024-09-11  **Leaving a Legacy in Media Art** The 29th Annual Conference of the Virtual Reality Society of Japan Organized Session
<https://conference.vrsj.org/ac2024/program/1E4.html>

Miscellaneous Experience (continued)

Social Contribution

2019  **Our Shurijo: Shuri Castle Digital Reconstruction Project**, Member,
<https://www.our-shurijo.org>

Skills

Design and prototyping	 Fusion 360, Rhinoceros + Grasshopper, ROS2, Blender, 3D printing (FDM, SLA), Arduino.
Programming	 Python, JavaScript, MATLAB, C++, Java.
RF simulation	 ADS, AWR, HFSS.
RF measurement skills	 Experience with Keysight, Tektronix, and R&S equipments.
Languages	 Korean (native), English (fluent, TOEIC score: 970), Japanese (fluent, JLPT score: 180).

Publications

Journal Articles

1. Yuki Kakui, Kota Araki, **Changyo Han**, Shogo Fukushima, and Takeshi Naemura. "Robust Recognition of Imperceptible QR Codes in Videos with a Dual-Camera Smartphone". In: *ITE Transactions on Media Technology and Applications* 13.1 (Jan. 2025), pp. 166–178. [DOI: 10.3169/mta.13.166](#).
2. Yu Shimada, **Changyo Han**, Ari Hautasaari, and Takeshi Naemura. "A Plane-Assisted Method for Efficient 3D Arrangement of Computer-Generated Objects". In: *The Journal of The Institute of Image Information and Television Engineers* 78.5 (July 2024), pp. 592–597. [DOI: 10.3169/itej.78.592](#).
3. Hwajin Lim and **Changyo Han**. "National borders transcended: the impact of geographical proximity on the growth of global innovation networks among cities in East Asia". In: *International Journal of Urban Sciences* 27.4 (Oct. 2023), pp. 570–598. [DOI: 10.1080/12265934.2021.1915854](#).
4. Kota Araki, Yuki Kakui, Ryotaro Tanaka, **Changyo Han**, Shogo Fukushima, and Takeshi Naemura. "A Color Vibration Method Robust to Interframe Difference for Embedding Imperceptible Markers in Videos". In: *The Journal of The Institute of Image Information and Television Engineers* 77.1 (2023), pp. 141–148. [DOI: 10.3169/itej.77.141](#).
5. **Changyo Han** and Takeshi Naemura. "BumpMarker: a 3D-printed tangible marker for simultaneous tagging, tracking, and weight measurement". In: *ITE Transactions on Media Technology and Applications* 7.1 (Jan. 2019), pp. 11–19. [DOI: 10.3169/mta.7.11](#).
6. **Changyo Han**, Minkyu Sung, Seung-Hyun Cho, Hwan Seok Chung, Sun Me Kim, and Jong Hyun Lee. "Performance Improvement of Multi-IFoF-Based Mobile Fronthaul Using Dispersion-Induced Distortion Mitigation With IF Optimization". In: *Journal of Lightwave Technology* 34.20 (Oct. 2016), pp. 4772–4778. [DOI: 10.1109/JLT.2016.2561297](#).
7. **Changyo Han**, Seung-Hyun Cho, Minkyu Sung, Hwan Seok Chung, and Jong Hyun Lee. "Clipping Distortion Suppression of Directly Modulated Multi-IF-over-Fiber Mobile Fronthaul Links Using Shunt Diode Predistorter". In: *ETRI Journal* 38.2 (Apr. 2016), pp. 227–234. [DOI: 10.4218/etrij.16.2515.0038](#).
8. Seung-Hyun Cho, **Changyo Han**, Hwan Seok Chung, and Jong Hyun Lee. "Demonstration of Mobile Fronthaul Test Bed Based on RoF Technology Supporting Two Frequency Assignments and 22 MIMO Antennas". In: *ETRI Journal* 37.6 (Dec. 2015), pp. 1055–1064. [DOI: 10.4218/etrij.15.0115.0146](#).
9. Minkyu Sung, **Changyo Han**, Seung-Hyun Cho, Hwan Seok Chung, and Jong Hyun Lee. "Improvement of the transmission performance in multi-IF-over-fiber mobile fronthaul by using tone-reservation technique". In: *Optics Express* 23.23 (Nov. 2015), p. 29615. [DOI: 10.1364/OE.23.029615](#).
10. Sun Hyok Chang, Hwan Seok Chung, Roland Ryf, Nicolas K. Fontaine, **Changyo Han**, Kyung Jun Park, Kwangjoon Kim, Jyung Chan Lee, Jong Hyun Lee, Byoung Yoon Kim, and Young Kie Kim. "Mode- and wavelength-division multiplexed transmission using all-fiber mode multiplexer based on mode selective couplers". In: *Optics Express* 23.6 (Mar. 2015), p. 7164. [DOI: 10.1364/OE.23.007164](#).

Preprints

1. Ryo Takahashi, **Changyo Han**, Wakako Yukita, John S. Ho, Takuya Sasatani, Akihito Noda, Tomoyuki Yokota, Takao Someya, and Yoshihiro Kawahara. *Full-body NFC: body-scale near-field sensor networks with machine-knittable meandered e-textiles*. Mar. 2025. [DOI: 10.48550/arXiv.2503.13240](https://doi.org/10.48550/arXiv.2503.13240).

International Conference Proceedings

1. **Changyo Han**, Yosuke Nakagawa, and Takeshi Naemura. “corobos: A Design for Mobile Robots Enabling Cooperative Transitions between Table and Wall Surfaces”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. CHI '25. New York, NY, USA: Association for Computing Machinery, Apr. 2025, pp. 1–16. [DOI: 10.1145/3706598.3713440](https://doi.org/10.1145/3706598.3713440).
2. Yuchi Yahagi, Rintaro Chujo, Yuga Harada, **Changyo Han**, Kohei Sugiyama, and Takeshi Naemura. “PaperWave: Listening to Research Papers as Conversational Podcasts Scripted by LLM”. In: *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. CHI EA '25. New York, NY, USA: Association for Computing Machinery, Apr. 2025, pp. 1–10. [DOI: 10.1145/3706599.3706664](https://doi.org/10.1145/3706599.3706664).
3. — “Temporal Perspectives on Exploring PaperWave: a Document-to-Audio Adaptation System powered by LLMs”. In: *Research Products and Time: When, For How Long, And Then What? Workshop in CHI 2025*. Yokohama, Japan, Apr. 2025, pp. 1–4.
4. **Changyo Han**, Cedric Caremel, Keiichiro Taniguchi, Hiroaki Murakami, Katsuya Koike, Koki Higashioka, Yoshihisa Toshima, Yasunori Akashi, and Yoshihiro Kawahara. “Acquisition and Visualization of HVAC Data by Autonomous Mobile Robots”. In: *Proceedings of the 30th CAADRIA Conference*. Vol. 3. Tokyo: CUMINCAD, Mar. 2025, pp. 29–38.
5. Yasunori Akashi, **Changyo Han**, and Takeshi Naemura. “MMM: Mid-air image Moving in and out of the Mirror with backward glance in the mirror”. In: *SIGGRAPH Asia 2024 Posters*. SA '24. New York, NY, USA: Association for Computing Machinery, Dec. 2024, pp. 1–2. [DOI: 10.1145/3681756.3697881](https://doi.org/10.1145/3681756.3697881).
6. Yuta Kibayashi, Kota Araki, Yasunori Akashi, **Changyo Han**, and Takeshi Naemura. “Quadrature Color Vibration Method for Improving Robustness of Imperceptible Markers Embedded in Video”. In: *2024 IEEE International Conference on Visual Communications and Image Processing (VCIP)*. Dec. 2024, pp. 1–5. [DOI: 10.1109/VCIP63160.2024.10849839](https://doi.org/10.1109/VCIP63160.2024.10849839).
7. Takafumi Morita, Rei Sakura, Kanon Aoyama, Tomomi Imamura, **Changyo Han**, and Yasuaki Kakehi. “Design and Fabrication for Dynamic Color-Changing on Curved 3D-Printed Surfaces”. In: *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. New York, NY, USA: ACM, May 2024, pp. 1–7. [DOI: 10.1145/3613905.3650742](https://doi.org/10.1145/3613905.3650742).
8. **Changyo Han**, Yosuke Nakagawa, and Takeshi Naemura. “Demonstrating Swarm Robots Capable of Cooperative Transitioning between Table and Wall”. In: *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. New York, NY, USA: ACM, Oct. 2023, pp. 1–4. [DOI: 10.1145/3586182.3615763](https://doi.org/10.1145/3586182.3615763).

9. Rei Sakura, **Changyo Han**, Yahui Lyu, Keisuke Watanabe, Ryosuke Yamamura, and Yasuaki Kakehi. "LattiSense: A 3D-Printable Resistive Deformation Sensor with Lattice Structures". In: *Proceedings of the 8th ACM Symposium on Computational Fabrication*. New York, NY, USA: ACM, Oct. 2023, pp. 1–14. [DOI: 10.1145/3623263.3623361](#).
10. Qunchao Zhou, Takuya Sasatani, **Changyo Han**, and Yoshihiro Kawahara. "Simultaneous Wireless Information and Power Transfer Technique Using Frequency-Modulated DC/DC Converters". In: *Asian Wireless Power Transfer Workshop (AWPT 2022)*. Kyoto, Dec. 2022, TU–4–O4.
11. Yuki Kakui, Kota Araki, **Changyo Han**, Shogo Fukushima, and Takeshi Naemura. "Using a Dual-Camera Smartphone to Recognize Imperceptible 2D Barcodes Embedded in Videos". In: *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology*. New York, NY, USA: ACM, Oct. 2022, pp. 1–3. [DOI: 10.1145/3526114.3558672](#).
12. Rei Sakura, **Changyo Han**, Keisuke Watanabe, Ryosuke Yamamura, and Yasuaki Kakehi. "Design of 3D-Printed Soft Sensors for Wire Management and Customized Softness". In: *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. New York, NY, USA: ACM, Apr. 2022, pp. 1–5. [DOI: 10.1145/3491101.3519906](#).
13. **Changyo Han**, Ryo Takahashi, Yuchi Yahagi, and Takeshi Naemura. "3D Printing Firm Inflatables with Internal Tethers". In: *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. New York, NY, USA: ACM, May 2021, pp. 1–7. [DOI: 10.1145/3411763.3451613](#).
14. Ikuo Kamei, **Changyo Han**, Takefumi Hiraki, Shogo Fukushima, and Takeshi Naemura. "CoVR: Co-located Virtual Reality Experience Sharing for Facilitating Joint Attention via Projected View of HMD Users". In: *SIGGRAPH Asia 2020 Emerging Technologies*. SA '20. New York, NY, USA: Association for Computing Machinery, Nov. 2020, pp. 1–2. [DOI: 10.1145/3415255.3422883](#).
15. Ryo Takahashi, Masaaki Fukumoto, **Changyo Han**, Takuya Sasatani, Yoshiaki Narusue, and Yoshihiro Kawahara. "TelemetRing: A Batteryless and Wireless Ring-shaped Keyboard using Passive Inductive Telemetry". In: *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*. UIST '20. New York, NY, USA: Association for Computing Machinery, Oct. 2020, pp. 1161–1168. [DOI: 10.1145/3379337.3415873](#).
16. **Changyo Han**, Katsufumi Matsui, and Takeshi Naemura. "ForceStamps: Fiducial Markers for Pressure-sensitive Touch Surfaces to Support Rapid Prototyping of Physical Control Interfaces". In: *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction - TEI '20*. New York, NY, USA: ACM, 2020, pp. 273–285. [DOI: 10.1145/3374920.3374924](#).
17. **Changyo Han**, Ryo Takahashi, Yuchi Yahagi, and Takeshi Naemura. "PneuModule: Using Inflatable Pin Arrays for Reconfigurable Physical Controls on Pressure-Sensitive Touch Surfaces". In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems - CHI '20*. New York, NY, USA: ACM, 2020, pp. 1–14. [DOI: 10.1145/3313831.3376838](#).
18. Keisuke Shiro, Ryotaro Miura, **Changyo Han**, and Jun Rekimoto. "An Intuitive Interface for Digital Synthesizer by Pseudo-intention Learning". In: *Proceedings of the 14th International Audio Mostly*

Conference: A Journey in Sound - AM '19. New York, NY, USA: ACM, Sept. 2019, pp. 39–44. [DOI: 10.1145/3356590.3356598](#).

19. **Changyo Han** and Takeshi Naemura. “BumpMarker: 3D printed marker for simultaneous tagging and weight measurement”. In: *2nd Annual ACM Symposium on Computational Fabrication*. poster presentation. Cambridge, MA, June 2018.
20. Seung-Hyun Cho, **Changyo Han**, Minkyu Sung, Hwan Seok Chung, Sun Me Kim, and Jong Hyun Lee. “Experimental Investigations of Uplink Transmission Performances in a Mobile Fronthaul based on IFoF Technique”. In: *International Conference on Information and Communication Technology Convergence (ICTC 2016)*. IEEE, Oct. 2016, pp. 772–774. [DOI: 10.1109/ICTC.2016.7763292](#).
21. Minkyu Sung, **Changyo Han**, Seung-Hyun Cho, Hwan Seok Chung, Sun Me Kim, and Jong Hyun Lee. “Bandwidth Efficient Transmission of 96 LTE-A Signals with 118-Gb/s CPRI-Equivalent Rate using 2-GHz Frequency Span and Intermixing Mitigation for Mobile Fronthaul”. In: *International Conference on Information and Communication Technology Convergence (ICTC 2016)*. IEEE, Oct. 2016, pp. 775–777. [DOI: 10.1109/ICTC.2016.7763293](#).
22. **Changyo Han**, Minkyu Sung, Seung-Hyun Cho, Hwan Seok Chung, Sun Me Kim, and Jong Hyun Lee. “Impact of Dispersion-Induced Second-Order Distortion in Multi-IFoF-based Mobile Fronthaul Link for C-RAN”. In: *Optical Fiber Communication Conference (OFC 2016)*. Washington, D.C.: OSA, Mar. 2016, Tu2B.4. [DOI: 10.1364/OFC.2016.Tu2B.4](#).
23. **Changyo Han**, Seung-Hyun Cho, Hwan Seok Chung, and Jong Hyun Lee. “Linearity Improvement of Directly-Modulated Multi-IF-over-Fibre LTE-A Mobile Fronthaul Link Using Shunt Diode Predistorter”. In: *European Conference on Optical Communication (ECOC 2015)*. IEEE, Sept. 2015, pp. 1–3. [DOI: 10.1109/ECOC.2015.7342016](#).
24. Seung-Hyun Cho, Hwan Seok Chung, **Changyo Han**, Sangsoo Lee, and Jong Hyun Lee. “Experimental Demonstrations of Next Generation Cost-Effective Mobile Fronthaul with IFoF technique”. In: *Optical Fiber Communication Conference (OFC 2015)*. Washington, D.C.: OSA, Mar. 2015, M2J.5. [DOI: 10.1364/OFC.2015.M2J.5](#).
25. Hwan Seok Chung, Seung-Hyun Cho, **Changyo Han**, Sangsoo Lee, Jyung Chan Lee, and Jong Hyun Lee. “Design of RoF based Mobile Fronthaul Link with Multi-IF Carrier for LTE/LTE-A Signal Transmission”. In: *2014 International Topical Meeting on Microwave Photonics and the 2014 9th Asia-Pacific Microwave Photonics Conference (MWP/APMP 2014)*. IEEE, Oct. 2014, pp. 303–306. [DOI: 10.1109/MWP.2014.6994559](#).
26. **Changyo Han**, Seung-Hyun Cho, Hwan Seok Chung, Sangsoo Lee, and Jong Hyun Lee. “Experimental Comparison of the Multi-IF Carrier Generation Methods in IF-over-Fiber System Using LTE Signals”. In: *2014 International Topical Meeting on Microwave Photonics and the 2014 9th Asia-Pacific Microwave Photonics Conference (MWP/APMP 2014)*. IEEE, Oct. 2014, pp. 311–314. [DOI: 10.1109/MWP.2014.6994561](#).
27. Seung-Hyun Cho, Hwan Seok Chung, **Changyo Han**, Sangsoo Lee, and Jong Hyun Lee. “Investigaitons of EVM Performance Degradations caused by Nonlienarity in Mobile Fronthaul Architecture based on

- IFoF technology". In: *12th International Conference on Optical Internet (COIN 2014)*. IEEE, Aug. 2014, pp. 1–2. [DOI: 10.1109/COIN.2014.6950590](#).
28. **Changyo Han**, Hwan Seok Chung, Sun Hyok Chang, Kwangjoon Kim, and Jong Hyun Lee. "Effect of rotational misalignment in phase-plate based mode multiplexer". In: *12th International Conference on Optical Internet (COIN 2014)*. IEEE, Aug. 2014, pp. 1–2. [DOI: 10.1109/COIN.2014.6950625](#).
 29. Hongbo Lu, Kazuro Kikuchi, Changyo Han, and Yojiro Mori. "Novel Polarization-diversity Scheme Based on Mutual Phase Conjugation for Fiber-nonlinearity Mitigation in Ultra-long Coherent Optical Transmission Systems". In: *European Conference and Exhibition on Optical Communication (ECOC 2013)*. IET, Sept. 2013, pp. 522–524. [DOI: 10.1049/cp.2013.1447](#).
 30. Yojiro Mori, **Changyo Han**, Hongbo Lu, and Kazuro Kikuchi. "Wavelength Demultiplexing of Nyquist WDM Signals under Large Frequency Offsets in Digital Coherent Receivers". In: *European Conference and Exhibition on Optical Communication (ECOC 2013)*. IET, Sept. 2013, pp. 122–124. [DOI: 10.1049/cp.2013.1313](#).
 31. **Changyo Han**, Koji Igarashi, and Kazuro Kikuchi. "Influence of Channel Misalignment of Time-interleaved DAC on Sensitivity Degradation in Coherent Optical Receivers". In: *Optical Fiber Communication Conference (OFC 2013)*. Washington, D.C.: OSA, Mar. 2013, OTh1F.2. [DOI: 10.1364/OFC.2013.OT1F.2](#).
 32. Yuki Tsukui, Hiroki Asada, **Changyo Han**, Kenichi Okada, and Akira Matsuzawa. "Area Reduction of Millimeter-Wave CMOS Amplifier Using Narrow Transmission Line". In: *Asia-Pacific Microwave Conference (APMC 2011)*. IEEE, Dec. 2011, pp. 797–800.
 33. Ryo Minami, **Changyo Han**, Kota Matsushita, Kenichi Okada, and Akira Matsuzawa. "Effect of Transmission Line Modeling Using Different De-embedding Methods". In: *European Microwave Conference 2011*. IEEE, Oct. 2011, pp. 381–384. [DOI: 10.23919/EuMC.2011.6101839](#).

Patents

1. Ryosuke Yamamura, Yasuaki Kakehi, **Changyo Han**, Rei Sakura, and Keisuke Watanabe. "Sensor and sensor manufacturing method". US20230349777A1 (US). 2023.
2. Seung-Hyun Cho, Jong Hyun Lee, Hwan Seok Chung, and **Changyo Han**. "Apparatuses and methods for transmitting and receiving control signal in analog radio-over-fiber (ROF)-based mobile fronthaul". US10027415 (US). 2018.
3. **Changyo Han**, Minkyu Sung, Jong Hyun Lee, Hwan Seok Chung, and Seung-Hyun Cho. "Optical signal transmission system and method of allocating center frequencies of intermediate frequency (IF) carriers for frequency division multiplexing (FDM) optical fiber link". US9998222B2 (US). 2018.
4. Sang Rok Moon and **Changyo Han**. "Mode-multiplexing control method, and transmission apparatus and reception apparatus for the same". US10027437B2 (US). 2018.
5. Minkyu Sung, Jong Hyun Lee, Hwan Seok Chung, Seung-Hyun Cho, and **Changyo Han**. "Analog optical transmission system using dispersion management technique". US10090928 (US). 2018.

6. Sun Hyok Chang, Kwangjoon Kim, Hyun Jae Lee, and **Changyo Han**. “Mode division multiplexed passive optical network (MDM-PON) apparatus, and transmission and reception method using the same”. US9787401B2 (US). Oct. 2017.
7. Hwan Seok Chung, Seung-Hyun Cho, Jong Hyun Lee, Sangsoo Lee, and **Changyo Han**. “Control apparatus and method for monitoring optical fiber link”. US9787395 (US). 2017.

Domestic Conference Proceedings

1. **Changyo Han**, Ryo Takahashi, Akihito Noda, and Yoshihiro Kawahara. “Roam beyond Battery: 長期的な屋内広範囲運用に向けた無線給電対応の小型移動ロボットネットワーク”. In: *Proceedings of the Robotics and Mechatronics Conference 2025*. June 2025, 2A1-J10.
2. Keita Inoue, **Changyo Han**, and Takeshi Naemura. “Preliminary Study on Acoustic Communication Combining Audible and Nearly-Inaudible Frequency Bands”. In: *メディアエクスペリエンス・バーチャル環境基礎研究会 (MVE)*. Tokyo, Japan: IEICE, June 2025.
3. Ryo Takahashi, **Changyo Han**, Wakako Yukita, Ho John S., Takuya Sasatani, Akihito Noda, Tomoyuki Yokota, Takeo Someya, and Yoshihiro Kawahara. “メアンダ形状の電子ニットによる全身電池レス無線センサネットワーク”. In: *研究報告ユビキタスコンピューティングシステム (UBI)*. Vol. 2025-UBI-86. May 2025, pp. 1–8.
4. Cedric Neergaard, Hiroaki Murakami, **Changyo Han**, and Yoshihiro Kawahara. “A Case Study on Determinants of Thermal Comfort: Insights into Air Conditioning and Environmental Factors”. In: *The 87th National Convention of IPSJ*. Osaka, Japan, Mar. 2025, 5ZC-09.
5. Ryoma Kanai, Yuta Kibayashi, Yasunori Akashi, **Changyo Han**, and Takeshi Naemura. “Extending Luminance Range and Maintaining Decoding Rate for Imperceptible Markers by Quadrature Color Vibration”. In: *映像メディア処理シンポジウム (IMPS 2024)*. Dec. 2024.
6. Yuchi Yahagi, Rintaro Chujo, Yuga Harada, **Changyo Han**, Kohei Sugiyama, and Takeshi Naemura. “Developing PaperWave: A System for Adapting Research Papers into Conversational Podcasts with LLMs”. In: *HCG シンポジウム 2024*. Dec. 2024, pp. 1–8.
7. Ryo Takahashi, Akihito Noda, **Changyo Han**, Tomoyuki Yokota, Takao Someya, and Yoshihiro Kawahara. “Design of full-body wireless charging and communication clothing for medical digital twin”. ja. In: *超知性ネットワーキングに関する分野横断型研究会 (RISING)*. Nov. 2024.
8. — “Design of full-body wireless charging and communication clothing for medical digital twin”. In: *革新的無線通信技術に関する横断型研究会 (MIKA)*. Oct. 2024.
9. Yasunori Akashi, **Changyo Han**, and Takeshi Naemura. “空中像光学系の水平視域拡大のための対称ミラー構造の設計”. In: *バーチャルリアリティ学会大会 2024*. Sept. 2024, 2E2-02.
10. Taichi Morimura, **Changyo Han**, and Takeshi Naemura. “corobos plus におけるエンタテインメントコンテンツのデザイン”. In: *エンタテインメントコンピューティングシンポジウム 2024 論文集*. Aug. 2024, pp. 505–509.

11. Yuchi Yahagi, Daisuke Kuramoto, **Changyo Han**, and Takeshi Naemura. “空中像をつくるワークショップにおいて参加者が直面する困難についての探索的検討”. In: 日本教育工学会全国大会. Mar. 2024, 3–So6B2.
12. Taichi Morimura, **Changyo Han**, and Takeshi Naemura. “corobos plus: 卓上・壁面・天井の間を移行する群ロボット”. In: インタラクシオン 2024. Feb. 2024, 1A11.
13. Ayako Yogo, Daisuke Kuramoto, **Changyo Han**, and Takeshi Naemura. “AIR-range Plus：テーブル上を動き回る複数の空中像とのインタラクシオン”. In: インタラクシオン 2024. Feb. 2024, p. 1C65.
14. — “机上を動き回る複数の空中像のインタラクティブシステム AIR-range Plus の基礎検討”. In: 映像情報メディア学会冬季大会 2023. Dec. 2023, pp. 21C–2.
15. Yuta Kibayashi, Kota Araki, **Changyo Han**, and Takeshi Naemura. “直交色振動不可視マーカにおける軟判定復号法の基礎検討”. In: 映像メディア処理シンポジウム (IMPS 2023). Nov. 2023, P2–07.
16. Rei Sakura, Mai Kagawa, **Changyo Han**, and Yasuaki Kakehi. “磁石を用いた組み換え可能なモジュール型ソフトセンサの提案”. In: Conference on 4D and Functional Fabrication 2023. Nov. 2023, OP–18.
17. Yasunori Akashi, **Changyo Han**, and Takeshi Naemura. “鏡に映り込む空中像と鏡に吸い込まれる空中像”. In: VR 大会 2023. Sept. 2023, pp. 1C2–02.
18. Yu Shimada, **Changyo Han**, Ari Hautasaari, and Takeshi Naemura. “CG オブジェクトの効率的配置のためのデスクトップ 3D インタフェースの開発”. In: 映像情報メディア学会冬季大会. Aug. 2023, pp. 22C–4.
19. Yasunori Akashi, Ayako Yogo, **Changyo Han**, and Takeshi Naemura. “再帰透過式空中像に対するプリズムシートを用いた迷光低減”. In: 3次元画像コンファレンス. July 2023, P–4.
20. Yousuke Nakagawa, **Changyo Han**, and Takeshi Naemura. “卓上と壁の間を移行する群ロボット corobos におけるアタッチメントの改良”. In: インタラクシオン 2023. Mar. 2023, 1B–46.
21. Kota Araki, Yuki Kakui, **Changyo Han**, and Takeshi Naemura. “Quadrature Color Vibration Method for Improving Robustness of Imperceptible Markers Embedded in Video”. In: MVE2023. 2023, pp. 7–12.
22. Kazuki Usui, Yuki Kakui, **Changyo Han**, and Takeshi Naemura. “Luminance-Based Data Representation for Improving Contrast in Pixel-level Visible Light Communication”. In: MVE2023. 2023, pp. 1–6.
23. Yasunori Akashi, Ayako Yogo, **Changyo Han**, and Takeshi Naemura. “再帰透過光学素子による空中像に対する最適な鑑賞方向の変換”. In: 映像情報メディア学会冬季大会 2022. 2022, p. 13Co2.
24. Minami Aramaki, Ari Hautasaari, **Changyo Han**, Shogo Fukushima, and Takeshi Naemura. “モード選択と手指動作入力を用いた対話的なアバターのモーション制御”. In: MVE2022. 2022, 信学技報, vol. 122, no. 74, MVE2022–2, pp. 7–12.

25. Yuki Kakui, Kota Araki, **Changyo Han**, Shogo Fukushima, and Takeshi Naemura. “動画に重畳した不可視マーカの頑健な認識 デュアルカメラによる同時撮影”. In: 第 21 回情報科学技術フォーラム (FIT2022). 2022, K-020.
26. Yousuke Nakagawa, **Changyo Han**, and Takeshi Naemura. “corobos: 卓上と壁の間を移行する群ロボット”. In: エンタテインメントコンピューティング 2022. 2022, pp. 106–109.
27. Yuchi Yahagi, Katsufumi Matsui, Ari Hautasaari, **Changyo Han**, Naho Tomiki, and Takeshi Naemura. “遠隔の初学者と経験者が共同で電子工作を行う製作ツールの提案”. In: インタラクシオン 2022. 2022, pp. 5D-05.
28. Kota Araki, Ryotaro Tanaka, Yuki Kakui, **Changyo Han**, Shogo Fukushima, and Takeshi Naemura. “動画像に不可視マーカを埋め込むためのフレーム間差分に頑健な色振動方式の基礎検討”. In: 映像メディア処理シンポジウム (IMPS 2021). 2021, P2-21.
29. Yuki Kakui, Ikuo Kamei, Ken Takaki, **Changyo Han**, Shogo Fukushima, and Takeshi Naemura. “可視光通信プロジェクトの高画質化のための光源制御による輝度補正”. In: MVE2021. 2021, vol. 121, no. 53, MVE2021-2, pp. 7-12.
30. Ryotaro Tanaka, Ikuo Kamei, Ken Takaki, **Changyo Han**, Shogo Fukushima, and Takeshi Naemura. “映像における色振動方式不可視マーカの読み取り可能距離の拡張”. In: MVE2021. 2021, vol. 121, no. 53, MVE2021-1, pp. 1-6.
31. **Changyo Han**, Ryo Takahashi, Yuchi Yahagi, and Takeshi Naemura. “PneuModule: 感圧タッチパッドとインフレータブルピンアレイを用いた再構成可能な物理インタフェース”. In: インタラクシオン 2020. Tokyo, Mar. 2020, 3B-17.
32. Ryo Takahashi, **Changyo Han**, Takuya Sasatani, Yoshiaki Narusue, Masaaki Fukumoto, Yoshihiro Kawahara, and Takeshi Naemura. “ワイヤレスな指輪型キーボードのバッテリーレス化手法”. In: 電子情報通信学会総合大会 2020. Mar. 2020, B-15-13.
33. Ikuo Kamei, **Changyo Han**, Takefumi Hiraki, Shogo Fukushima, and Takeshi Naemura. “HMD 視点映像のプロジェクト投影による VR 体験の共有”. In: VR 大会 2020. 2020, 2B1-4, OVE:A11-(3).
34. Saho Yamaguchi, **Changyo Han**, and Takeshi Naemura. “映像コンテンツとロボットの連携に向けた非可聴音声への信号埋め込み手法の検討 Embedding Signals for Robot-Video Communication Using Inaudible Frequency Sounds”. In: MVE2018. 2018, pp. 1-6.
35. Sun Hyok Chang, Hwan Seok Chung, **Changyo Han**, Kwangjoon Kim, Jyung Chan Lee, and Jong Hyun Lee. “3 개의 모드를 이용한 모드 분할 다중 및 파장 분할 다중 광전송”. In: 광전자 및 광통신 학술회의 (COOC 2015). 2015, T2A-I2.
36. **Changyo Han**, Hwan Seok Chung, Sun Hyok Chang, Kwangjoon Kim, and Jong Hyun Lee. “6x6 MIMO 신호 처리를 이용한 위상 판 기반 모드 다중화기의 성능 평가”. In: 광전자 및 광통신 학술회의 (COOC 2015). 2015, T1A-5.

37. Andres C. Najarro, Sung Man Kim, **Changyo Han**, Seung-Hyun Cho, Hwan Seok Chung, and Jong Hyun Lee. "Nonlinear Compensation based on Predistortion for Radio over Fiber Systems". In: 광전자 및 광통신 학술회의 (COOC 2015). 2015, W1A-I3.
38. Hwan Seok Chung, Sun Hyok Chang, **Changyo Han**, Kwangjoon Kim, Jyung Chan Lee, and Jong Hyun Lee. "DP-16QAM 과 4x4 MIMO 신호처리를 이용한 모드분할다중화 광 송수신실험 ". In: 광전자 및 광통신 학술회의 (COOC 2014). 2014, T2A-I2.
39. **Changyo Han**, Hwan Seok Chung, Sun Hyok Chang, and Kwangjoon Kim. "위상 판을 이용한 모드 다중화기의 구현 ". In: 광전자 및 광통신 학술회의 (COOC 2014). 2014, T2A-I3.
40. Daiki Hiraoka, **Changyo Han**, and Kazuro Kikuchi. "Suppression of Interband-Crosstalk Penalty of Nyquist-WDM Signals using Adaptive FIR Filters". In: 電子情報通信学会総合大会 2013. Mar. 2013, B-10-51.
41. **Changyo Han**, Koji Igarashi, and Kazuro Kikuchi. "Influence of channel misalignment of time-interleaved DAC on sensitivity degradation in coherent receivers". In: 電子情報通信学会光通信システム研究会. Aug. 2012, OCS-2012-22.
42. **Changyo Han**, Hiroki Asada, Kota Matsushita, Keigo Han, Kenichi Okada, and Akira Matsuzawa. "Investigation on CMOS Interconnects for High Frequency Applications". In: 電子情報通信学会ソサイエティ大会. Sept. 2010.

References

Available on Request